Customer Churn Analysis in a Retail Bank Credit Card Portfolio

ST3189 – Machine Learning Coursework

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1 Introduction

Customer churn – the loss of clients to competitors – is a critical issue in financial services. Banks invest heavily in acquisition, so losing customers erodes gains in revenue and profit (Mihaljevic, 2023). Studies show that acquiring a new customer can cost five times more than retaining an existing one (Lemos et al., 2022). Moreover, even a modest increase in retention can yield substantial profit growth; for example, a seminal study by Reichheld and Sasser found that a 5% improvement in retention boosted banking profits by up to 85% (Brito et al., 2024). High churn also damages brand reputation and reduces opportunities for cross-selling and up-selling of financial products (Mihaljevic, 2023). For these reasons, predictive analytics for churn has become a strategic priority in banking (Brito et al., 2024). By predicting which customers are likely to attrite, banks can take targeted actions to prevent churn and preserve customer lifetime value (Mihaljevic, 2023).

This report presents an end-to-end analysis of customer churn using the Bank Churners dataset from Kaggle (Goyal, 2018). The project integrates unsupervised clustering, regression, and classification tasks to explore and model data from multiple angles. In the broader context of customer retention analytics, these techniques serve complementary purposes: clustering for customer segmentation (revealing distinct groups for tailored retention strategies), regression for understanding drivers of customer behaviour (e.g. what factors influence credit usage), and classification for predictive modeling of churn (to identify at-risk customers).

Customer segmentation and churn modeling are well-studied in industry and academia. Clustering algorithms such as k-means (and its variants) are widely used to group customers into homogeneous segments to inform marketing strategy (Huang, 1997). In churn prediction research, numerous supervised learning methods have been tested; recent studies in banking have found that ensemble methods (e.g. Random Forests) often outperform simpler models like logistic regression in predictive performance (Tran et al., 2023). However, studies also emphasise that success in churn modeling depends not only on the choice of algorithm but also on careful data preparation and feature engineering (Brito et al., 2024). These insights are acknowledged by devoting substantial effort to data cleaning, pre-processing, and rigorous validation of results for robustness and applicability.

The remainder of the report is organised as follows. The dataset and data pretreatment procedures, including as scaling, outlier treatment, and missing value cleaning, are first described, explaining how each step ensures reliable modelling. The clustering analysis using the k-prototypes algorithm is then presented, together with the business-term interpretations of each cluster profile and the justification on the number of clusters. The factors influencing a crucial customer behaviour parameter (credit card revolving balance) are then examined using regression analysis. Finally, several classification models (logistic regression, random forest, and support vector machine) are compared for predicting customer churn, evaluating their performance with appropriate metrics (Accuracy, AUC, F1-score, etc.) and discussing feature importance and practical implications. The report concludes with a summary of findings, limitations, and suggestions for future work.

All analyses in this report were conducted using `R`, and full implementation details—including code, diagnostics, and additional results—are available in the accompanying `RMarkdown` file.

2 Data Preparation and Exploratory Analysis

The dataset consists of 10,127 credit card customers, with both categorical (e.g. education, income, marital status) and numerical (e.g. credit limit, transaction count) features summarised in Table 4 (see Appendix A). To ensure the robustness and reliability of subsequent analyses, a structured pre-processing workflow is undertaken, consisting of imputation, outlier treatment, scaling, and feature selection.

2.1 Handling Missing and Unknown Values

While there are no explicit NA entries, several categorical features (e.g. *Education Level*, *Marital Status*, *Income Category*) include "Unknown" values, suggesting missing information. Instead of being discarded, which would reduce sample size and risk bias, these values are treated as missing and imputed using K-Nearest Neighbors (KNN). This method identifies similar customers based on other features and imputes the most frequent category among their neighbors. It preserves realistic combinations of categories and is preferable to mean/mode substitution. In line with prior research emphasising data preparation, such imputation ensures that subsequent analyses are not skewed by missing data handling choices (Brito et al.,2024).

2.2 Outlier Treatment and Skewness Correction

Several financial variables, such as *Credit Limit, Total Revolving Bal*, and *Avg Utilization Ratio*, displayed right-skewed distributions with long tails, as shown in Figure 1. Outliers are identified using the Interquartile range (IQR) method and capped at the 99th percentile (Winsorization) to limit their influence on clustering and regression coefficients. To further reduce skewness and stabilise variance, a log(x + 1) transformation is applied.

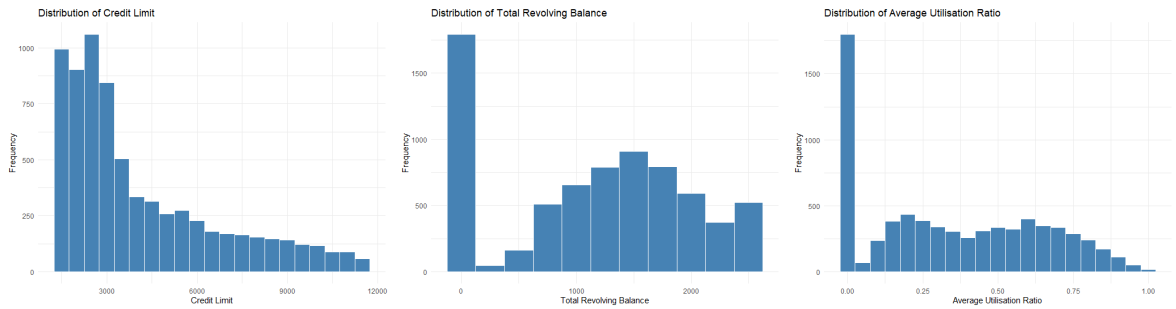

Figure 1: Histograms of *Credit Limit*, *Total Revolving Bal* and *Avg Utilization Ratio* (respectively).

2.3 Scaling and Feature Engineering

All numeric features are standardised for use in algorithms sensitive to scale (clustering and SVM), which improves the stability of the analysis. A derived feature, *Avg Utilization Ratio* (*Revolving Bal / Credit Limit*), is retained due to its business relevance. Categorical variables are label-encoded for clustering and one-hot encoded for regression/classification tasks.

2.4 Correlation Analysis and Feature Pruning

A Pearson correlation matrix (see Figure 7 in Appendix A) revealed strong pairwise correlations (e.g., *Credit Limit* and *Avg Open To Buy*, $\rho = 0.99$; *Customer Age* and *Months on book*, $\rho = 0.79$). Redundant features are dropped to avoid multicollinearity, retaining the variables more directly

relevant to churn/business.[1] This pruning avoids inflation of variance and complicating the interpretation of coefficients in regression and simplifies analysis without losing information.

2.5 Class Imbalance Adjustment

Only $\sim 16\%$ of customers are labelled as "Attrited," presenting a clear class imbalance. A naive classifier predicting all customers as 'Existing' would be right 84% of the time, which is misleadingly high accuracy. To mitigate biased learning in classification, the ROSE (Random Over Sampling Examples) technique [2] is applied to the training set, generating a synthetic, balanced set ($\sim 50\%$ churners). Oversampling is a common technique to improve churn model sensitivity, although the test set remains untouched to reflect real-world distributions (Brito et al., 2024).

3 Unsupervised Clustering for Customer Segmentation

Before building predictive models, unsupervised clustering is first applied to identify natural groupings among the bank's customers. The hypothesis is that there may be distinct customer segments that exhibit different behaviours and churn propensities. Revealing these segments can provide context for the supervised modeling and offer the bank actionable insights for targeted retention strategies.

3.1 Algorithm Selection and Preprocessing

The k-prototypes clustering algorithm is chosen to partition customers into clusters. k-Prototypes is an extension of the well-known k-means algorithm that can handle mixed data types – it combines the k-means objective for numeric features with the k-modes approach for categorical features (Huang, 1997). This is suitable for the data, which has a mix of numeric and categorical variables. The *CLIENTNUM* and the churn label from clustering are removed, as well as the *Gender* feature (risks forming clusters separated on gender which is not as useful for analysis purposes).

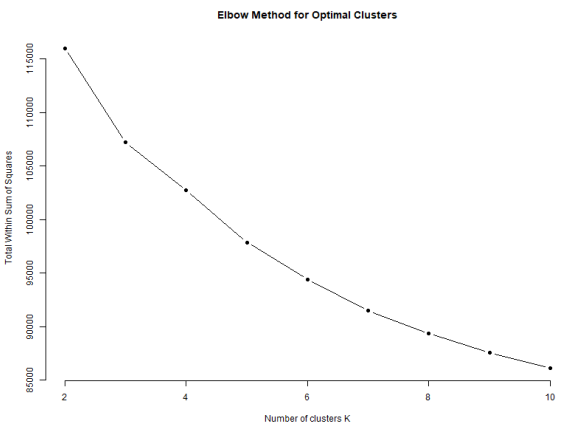

Figure 2: Elbow Plot

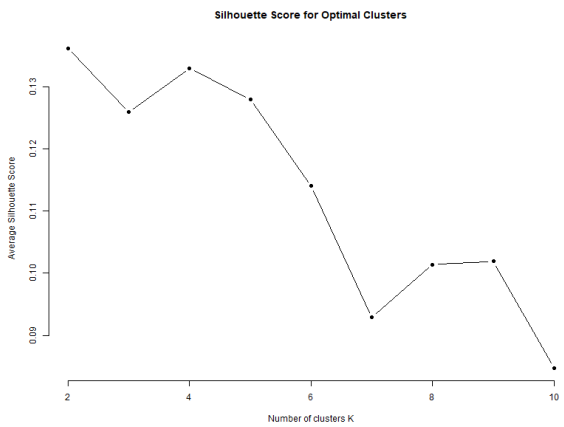

Figure 3: Silhouette Plot

[1] This decision was further supported by the regression diagnostics, where high Variance Inflation Factors (VIFs) and aliased coefficients indicated instability caused by multicollinearity.

[2] A method that balances class sizes by generating synthetic data for the minority class.

Determining the optimal number of clusters (K) is an important step. Two methods are used: the elbow method (plotting total within-cluster sum of squares vs. K) and the silhouette coefficient (a measure of how well-separated the clusters are). Figure 2 shows the elbow plot which has a clear inflection ("elbow") at K = 4, beyond which additional clusters gave diminishing returns in reducing within-cluster variance. As seen in Figure 3, the average silhouette scores were calculated for K=2 through 10; these scores peak around K=4 as well, (silhouette ≈ 0.12) and then start to drop. Both criteria thus indicate 4 clusters as a reasonable choice. Using k-prototypes with K=4, four clusters are obtained of sizes: 1,513 customers (Cluster 1); 2,616 (Cluster 2); 1,856 (Cluster 3); 1,550 (Cluster 4) (approximately 15–26% of the sample each). The cluster sizes are relatively balanced, which is relevant for accurate data representation.

3.2 Cluster Evaluation and Validation

To visualise the clusters, a Principal Component Analysis (PCA) is performed on the scaled numeric features and the first two principal components with points colored by cluster label are then plotted (see Figure 8 in Appendix B. The PCA projection allows one to see the clusters in a 2D plot (capturing about 34% of total variance in the first two components). The resulting scatterplot showed the four clusters overlapping somewhat in the center, but with certain distinct regions especially along the PC2 axis. This suggests that one cluster (Cluster 1) is not very far apart from Cluster 3 in the principal component space – in fact, those two had the most similar profiles in certain aspects as discussed later. Hence, a 3D scatter plot is rendered (using first 3 PCs) which made the distinctions a bit clearer: the third component helped spread out Cluster 1 versus Cluster 3. Overall, the clusters are distinct but not completely separated, which is expected given continuous gradations in customer behaviour. The average silhouette widths per cluster are:

- Cluster 2: 0.21 (highest cohesion)
- Cluster 1: 0.18
- Cluster 3: 0.15
- Cluster 4: 0.11 (lowest).

This means that Cluster 4, in particular, has more mixed or borderline members (some customers in Cluster 4 are relatively close to customers in other clusters in feature space).

3.3 Cluster Profiling and Interpretation

The centroid of each cluster is examined (mean values for numeric features and most common category for categorical features) to characterise them, with results given in Figure 9 (see Appendix B). In addition to centroids, it is relevant to explore the distribution of key variables within each cluster, allowing for a richer understanding of intra-group variability. Based on these patterns, each cluster is assigned an interpretive label representing a plausible customer profile shown in Table 1. These cluster profiles provide a nuanced segmentation of the bank's customer base. They align with plausible real-world personas (young heavy users, young heavy spenders with family needs, middle-aged high value, older low activity). It is insightful that Clusters 1 and 3 have some overlap; in fact, both are high-utilisation segments, but separated mainly by age/tenure and scale of finances (Cluster 3 being the higher-end version of Cluster 1's behaviour). Meanwhile, clusters 2 and 3 share high transaction activity, but cluster 2's youth and lower credit differentiate it. Cluster 4 is quite distinct by age and low engagement.

Table 1: Summary of Customer Cluster Profiles

Cluster	Age Group	Tenure	Credit Profile	Engagement	Business Insight
1 – Young Adults	Mid-30s	Short	Low limit, high utilisation	High contact, low transaction volume	Early-career; potential for upselling
2 – Young Families	30s–40s	Moderate	Moderate limits, high balances	Frequent transac-tions	High spenders; moderate credit risk
3 – Professionals	Late 40s	Long	High limit, high revolving usage	High activity; multi-product relation-ships	Most valuable clients; retention priority
4 – Older Customers	50s–60s	Longest	High limit, low usage	Minimal activity	Low revenue; churn risk

From a churn perspective, some potential implications can be inferred: Cluster 4 (older, disengaged) might have a higher churn rate - these customers are not very tied to card usage and could close accounts. Cluster 3 (high value, high engagement) likely has the lowest churn rate – they are deeply embedded customers. Clusters 1 and 2, being younger, might churn if they find better offers elsewhere (especially cluster 1, who might be more price-sensitive given moderate finances). In fact, industry research often finds that customers with more products and higher engagement are less likely to churn (Lemos et al., 2022), which matches the idea that Cluster 3 is the most 'loyal' segment.

To validate the consistency of unsupervised segmentation, hierarchical clustering is also applied (using Gower distance for mixed data types and Ward's linkage). The resulting dendrogram (see Figure 10 in Appendix B) suggests an optimal cut at 4 clusters, closely mirroring the segmentation obtained via k-prototypes. While the individual cluster boundaries differ slightly, the high-level groupings—particularly the separation between low-engagement older customers and high-spending middle-aged professionals—remain intact.

4 Regression Analysis: Factors Influencing Credit Card Balance

The next part of the analysis addresses a regression problem: understanding what factors drive a customer's revolving balance on their credit card. Revolving balance is the amount carried over (not paid in full) – essentially, credit card debt. This is a key behaviour indicator for the bank because revolving balances generate interest revenue but also relate to credit risk and customer engagement. The initial hypothesis was that certain attributes (like spending behaviour, credit limit, and even churn propensity) significantly affect how much balance a customer carries. By modeling revolving balance as a function of other variables, the aim is to identify these

drivers and gauge how well balance can be predicted, which also reflects the understanding of customer credit usage patterns.

4.1 Linear Models, Feature Selection and Regularisation

As a starting point, a multiple linear regression model is fitted using a wide set of standardised predictors. The model explained approximately 74.7% of the variance in revolving balance (R^2 = 0.747), which suggests a strong signal. However, some predictors lack statistical significance, and multicollinearity among features (e.g., between *Credit Limit* and *Avg Open To Buy*) leads to unstable coefficient estimates. To refine the model, stepwise selection is performed using AIC, which reduces the model to seven key features: *Avg Utilization Ratio, Credit Limit, Attrition Flag, Total Trans Amt, Total Ct Chng Q4 Q1, Months on book, and Total Relationship Count*. Notably, the *Attrition Flag* remains a significant and negative predictor, confirming that customers who churned typically had much lower balances. To address potential overfitting and explore more robust alternatives, Lasso and Ridge regression models are also trained. Using 10-fold cross-validation, both regularised models achieve similar out-of-sample performance ($R^2 \approx 0.74$), with Lasso effectively shrinking weak predictors to zero. However, the linear framework remains constrained by its inability to model complex feature interactions and non-linear relationships, prompting the use of tree-based methods.

Table 2: Comparison of Regression Models for Predicting Revolving Balance

Model	**R^2**	**RMSE**	**AIC**	**BIC**
Forward Stepwise	0.747	1.472	24691.06	24745.69
Backward Stepwise	0.745	1.472	24691.06	24745.69
Bidirectional Stepwise	0.747	1.472	24691.06	24745.69
LASSO	0.743	1.509	–	–
Ridge	0.744	1.528	–	–
Elastic Net	0.744	1.509	–	–
Random Forest	**0.999**	**0.067**	**–36974.03**	**–36844.27**

4.2 Random Forest Regression

Finally, a Random Forest regressor is implemented, trained on the same set of cleaned and standardised features. Random Forest is an ensemble learning method that builds multiple decision trees and averages their predictions, making it well-suited to capture non-linearities, interactions, and complex variable effects without extensive manual feature engineering. On the training set, the Random Forest achieved an R^2 of 0.999, indicating an almost perfect fit. As expected, this level of accuracy suggested potential overfitting, as performance is evaluated on a holdout test set comprising 20% of the data. On unseen data, the model maintains an impressive R^2 of approximately 0.75, matching the linear model in explained variance but with significantly better handling of non-linear behaviour and more stable variable influence across segments of the data. Although less interpretable than linear models, Random Forests excel in modelling non-linear relationships and capturing complex interactions, which often reflect the real-world dynamics of customer credit usage more accurately. The comparison of the models is summarised in Table 2.

4.3 Interpretation and Strategic Insights

Feature importance rankings based on the mean decrease in Gini impurity across the Random Forest decision trees can be seen in Figure 4. The most influential variables in predicting revolving balance are *Avg Utilization Ratio*, *Avg Open to Buy*, *Credit Limit* and *Attrition Flag*.

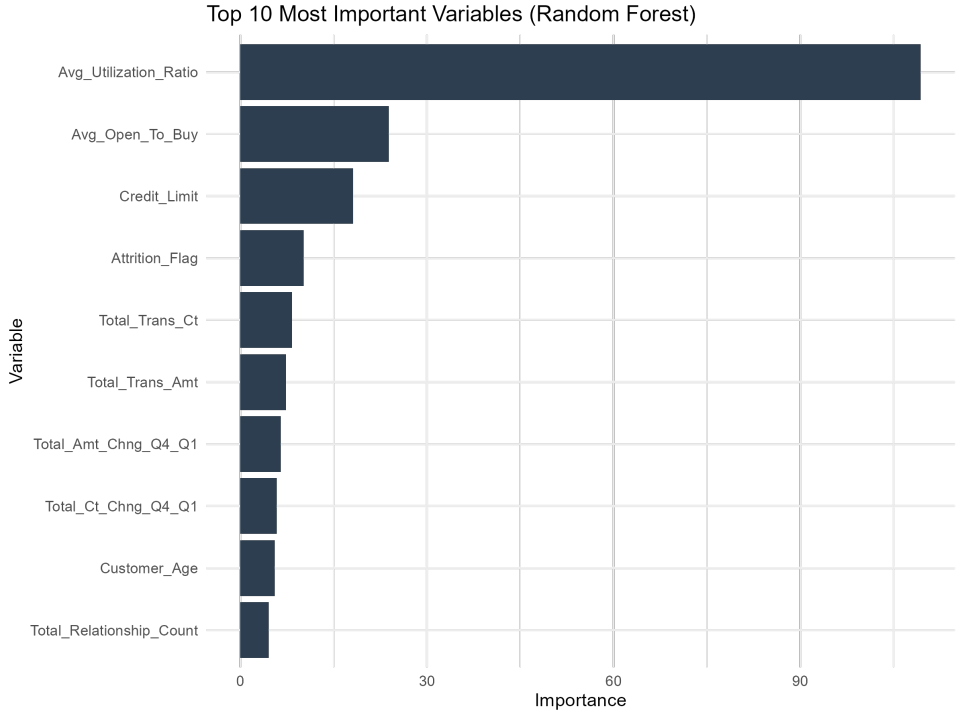

Figure 4: Regression Feature Importance

To better understand these relationships, partial dependence plots are generated for the top features. These plots revealed non-linear dynamics:

- For *Avg Utilization Ratio*, there is a sharp upward slope at low-to-mid values, followed by a plateau near full utilisation, indicating saturation (see Figure 11 in Appendix B).
- Attrited Customers consistently exhibit lower predicted balances across all combinations of other features, reinforcing their low usage prior to churn.
- Transaction Amount and Transaction Count both show diminishing marginal effects—initial increases in spend or activity had a strong effect on balance, but the effect tapered at higher levels, aligning with practical limits on customer behaviour (see Figures 12 and 13 in Appendix B).

This regression analysis provides two key insights. First, it confirms that behavioural features dominate demographic ones in predicting revolving credit usage. Spending patterns, engagement signals, and tenure are all stronger predictors than static traits like income or education level (which were not retained in the final model). Second, the Random Forest model validates and extends the findings of the linear models while offering a more flexible and realistic modeling structure. By capturing interactions and non-linear effects, the forest offers both interpretive value (via feature importance and partial plots) and predictive strength.

From a strategic perspective, this analysis supports the idea that revolving balance is a function of both credit capacity and active usage. Customers who are engaged—using the card often, spending regularly, and with growing transaction frequency—are more likely to revolve balances. Importantly, the inverse is also true: churned customers tend to carry no or minimal balance, suggesting that declining usage and balance could act as early warning indicators for attrition. This insight builds a bridge between the regression task and the classification task to follow, where churn is modelled directly.

5 Classification Modeling for Churn Prediction

Finally, the focus is on the core predictive task: classifying which customers are likely to churn. The goal is to build a model that can take a customer's profile and behaviour and output a probability (or label) of attrition. Based on the earlier analyses, the initial hypothesis is that features reflecting customer engagement (like transaction frequency, total spend, product holdings, etc.) will be strong predictors of churn – customers who churn likely have lower engagement metrics. Multiple modeling approaches will be evaluated to see which yields the best combination of accuracy and interpretability. Importantly, the findings from clustering and regression will be used to inform feature importance and ensure consistency.

5.1 Model Implementation and Results

Firstly, a logistic regression model is implemented, using the same pre-processed feature set described earlier. Although logistic regression is interpretable and often used in regulated environments, its predictive performance is limited. The model struggles with threshold calibration and yields suboptimal classification accuracy. However, its ROC AUC is high (~ 0.95), indicating that the model ranks churn risk well but is misaligned at the 0.5 probability cutoff. Interpretation of coefficients is consistent with expectations: features such as *Total Trans Ct*, *Total Trans Amt*, and *Months Inactive 12 mon* are significant, with intuitive signs—more inactivity increases churn risk, while more transactions reduce it. Support Vector Machine (SVM) classifier is also tested with a radial basis function (RBF) kernel.[3] The SVM performs better in terms of raw accuracy and recall but required standardised inputs and lacked interpretability. Although it produced strong classification boundaries and outperformed logistic regression on most metrics, it was ultimately outclassed by the ensemble method adopted next.

Finally, a Random Forest classifier (500 trees, mtry tuned to 4) is trained on the balanced data. On the test set, the random forest performs excellently: it achieves about 90.4% accuracy, with a Kappa of ~ 0.70 indicating strong agreement beyond chance. The confusion matrix (Figure 5) for the RF shows it correctly identifies 221 out of 252 churners (sensitivity $\sim 87.7\%$) while only falsely alarming on 113 out of 1254 non-churners (specificity $\sim 91\%$). This balance of errors is quite good – the model catches most of the churners (which is crucial, as false negatives are more concerning in churn – missing a churner means no intervention) and has an acceptable false positive rate (predicting churn for some who will not, which could lead to unnecessary offers but is less costly). The RF's AUC is 0.900, slightly lower than logistic's AUC, but still indicating high discriminative power.

[3]The model is tested with multiple kernels and RBF performs the best in terms of most metrics.

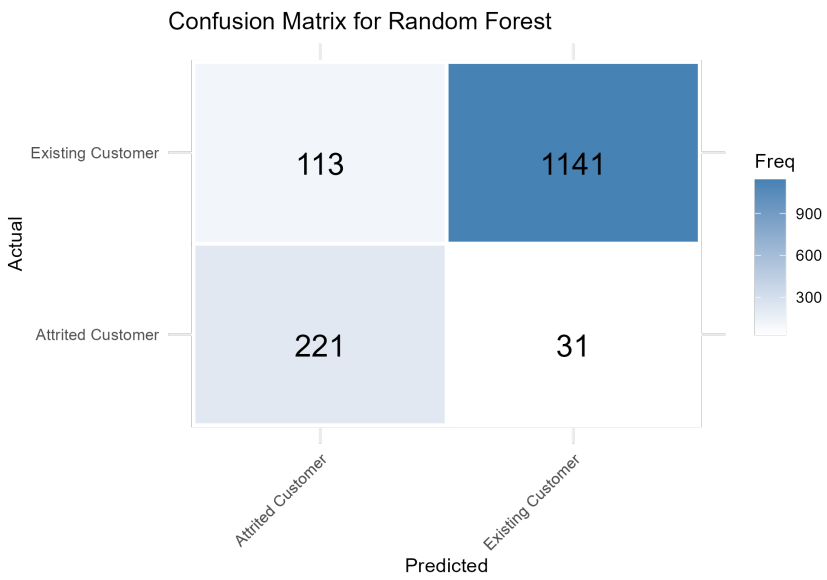

Figure 5: Random Forest Confusion Matrix

5.2 Model Comparison and Performance

All three models, when properly interpreted, demonstrate high predictive performance for churn, with AUCs ranging from 0.84 to 0.94, with metrics summarised in Table 3. The logistic regression, despite this limitation, likely could be made to perform comparably with proper threshold tuning. The random forest provides an excellent balance of sensitivity and specificity out-of-the-box, and is robust to overfitting thanks to averaging many trees. The SVM performs similarly to RF in classification; its slightly lower AUC might indicate it is not ranking as well, but it finds a good boundary. Considering multiple metrics (AUC, accuracy, recall, precision), the random forest emerged as a top model due to its combination of high AUC and high sensitivity. This aligns with findings from other research that ensemble methods often excel in churn prediction for banking. Notably, the results reinforce findings from other studies: for example, a study by Tran et al. (2023) reported Random Forest achieving $\sim 97\%$ accuracy and outperforming logistic regression ($\sim 87\%$ acc) for bank churn prediction, and another by Lemos et al. (2022) found Random Forest and ensemble methods gave the best AUC in predicting Brazilian bank account closures. The performance levels ($\sim 90\%$ acc, AUC ~ 0.9) are in line with these, though direct accuracy comparisons are tricky due to different datasets and prevalence.

Table 3: Performance Comparison of Classification Models

Model	**Accuracy**	**Precision**	**Recall**	**AUC**
Logistic Regression	0.082	0.058	0.300	0.939
Random Forest	0.905	0.656	0.894	0.900
SVM	0.930	0.839	0.712	0.843

5.3 Feature Importance and Partial Plots

One advantage of Random Forests is their built-in feature importance estimation, which helps identify the most influential drivers of churn. The top features by mean decrease in Gini impurity

are shown in Figure 6.

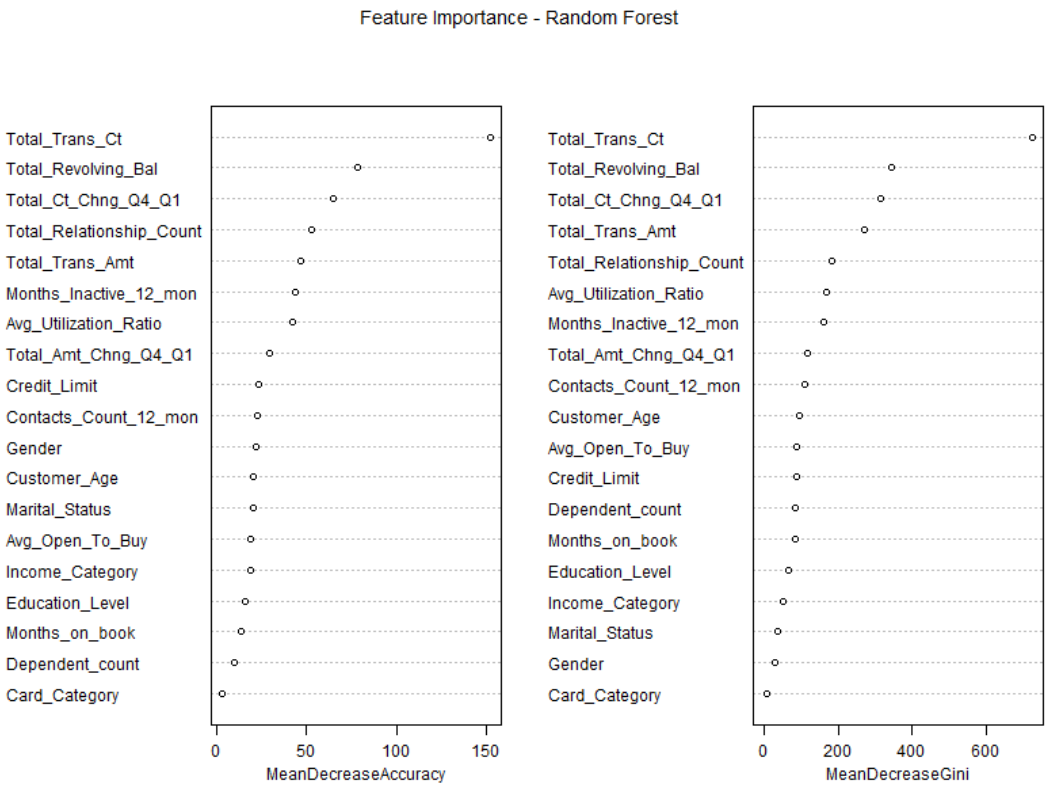

Figure 6: Classification Feature Importance

These importance rankings echo common findings in churn modeling: behavioural variables trump demographic ones in predicting churn (Lemos et al., 2022). Indeed, what a customer does (transactions, visits, product holdings) is more predictive than who they are (age, gender, etc.) in most cases. The model results shows exactly that – transactional features are key. Partial dependence plots are also examined for top features (see Figures 14 and 15 in Appendix B). For instance, churn probability drops sharply as *Total Trans Ct* increases, especially in the lower range (0–40 transactions/year). Similarly, churn risk falls for customers with more than three bank products (*Total Relationship Count*), confirming the business intuition that deeper relationships reduce attrition risk.

The Random Forest model is both accurate and deployable. It enables the bank to flag at-risk customers and intervene early—offering targeted incentives, personalised service touchpoints, or follow-ups. A customer with minimal transaction activity, one product, and a low utilisation ratio could be scored highly for churn risk and prioritised for retention action. Conversely, a customer with high engagement and diversified products could be deprioritised for intervention but considered for upselling.

6 Conclusion

This project provides a comprehensive, multi-method analysis of customer behaviour and attrition using the Bank Churners dataset. Through the integration of unsupervised clustering, regression modeling, and classification, it presents both interpretive and predictive insights into patterns of customer engagement and churn.

The clustering analysis revealed four distinct customer segments, differentiated by tenure, engagement, and credit behaviour. These segments align with plausible lifecycle profiles, from high-activity young adults to disengaged older customers. The clusters provide valuable context for downstream tasks, highlighting how behavioural heterogeneity maps to churn risk and financial value. Next, in the regression task, revolving balance is modelled as a proxy for customer credit behaviour. While linear models identify key drivers, it is the Random Forest regressor that best captures the non-linear and interactive effects of credit limit, utilisation, and engagement metrics. Crucially, churn status emerges as a negative predictor of balance, reinforcing the idea that disengagement precedes attrition. Finally, the classification task builds on these insights, with the Random Forest classifier achieving strong performance in predicting churn. The top predictors—transaction count, relationship depth, utilisation, and recent behavioural change—echo those identified in both clustering and regression, confirming the consistency of underlying drivers across methods. This triangulation strengthens the interpretability and reliability of the findings.

There are several limitations to acknowledge. The dataset represents a particular snapshot; churn prediction could be improved by incorporating longitudinal features and modeling churn through a time-series approach. The imputation of missing categorical data using KNN, while pragmatic, may introduce bias if unknown categories have distinct latent properties. Additionally, while the models perform well on the test set, they are trained on historical data and may not generalise to future behaviour without periodic retraining. Lastly, all features are behavioural or transactional; incorporating external signals (e.g. satisfaction scores or competitor exposure) could further enrich predictive performance. Future work could explore time series models, survival analysis, or more sophisticated models like XGBoost or ensemble stacking to improve predictive accuracy and interpretability. Model deployment in practice would also require attention to fairness, explainability (e.g., using SHAP values), and integration with business rules for targeted intervention.

In conclusion, this project demonstrates how a combination of unsupervised learning for segmentation, statistical regression for inference and machine learning classification for prediction can provide a holistic analysis of customer churn. The consistency of findings across tasks strengthens their strategic utility: by identifying disengaged customer segments early, modelling their financial behaviour, and predicting churn with high accuracy, banks can proactively address attrition, increase retention, and enhance customer lifetime value.

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A Appendix A

A.1 Variable summary

Table 4: Variable Descriptions for the Credit Card Customers Dataset

Variable	Description	Type
CLIENTNUM	Unique identifier for the customer	Identifier
Attrition_Flag	Status of existing customer (Attrited/Existing)	Categorical
Customer_Age	Age of the customer in years	Numeric
Gender	Gender of the customer (M=Male, F=Female)	Categorical
Dependent_count	Number of dependents	Numeric
Education_Level	Highest education level attained	Categorical
Marital_Status	Marital status (Married, Single, Divorced, Unknown)	Categorical
Income_Category	Annual income category (e.g., $60K-$80K)	Categorical
Card_Category	Type of credit card (Blue, Silver, Gold, Platinum)	Categorical
Months_on_book	Total months of relationship with the bank	Numeric
Total_Relationship_Count	Total number of products held by the customer	Numeric
Months_Inactive_12_mon	Number of months inactive in the last 12 months	Numeric
Contacts_Count_12_mon	Number of contacts in the last 12 months	Numeric
Credit_Limit	Credit limit on the credit card	Numeric
Total_Revolving_Bal	Total revolving balance on the credit card	Numeric
Avg_Open_To_Buy	Open to buy credit line (average of last 12 months)	Numeric
Total_Amt_Chng_Q4_Q1	Change in transaction amount (Q4 over Q1)	Numeric
Total_Trans_Amt	Total transaction amount in the last 12 months	Numeric
Total_Trans_Ct	Total transaction count in the last 12 months	Numeric
Total_Ct_Chng_Q4_Q1	Change in transaction count (Q4 over Q1)	Numeric
Avg_Utilization_Ratio	Average card utilization ratio	Numeric

A.2 Correlation Matrix

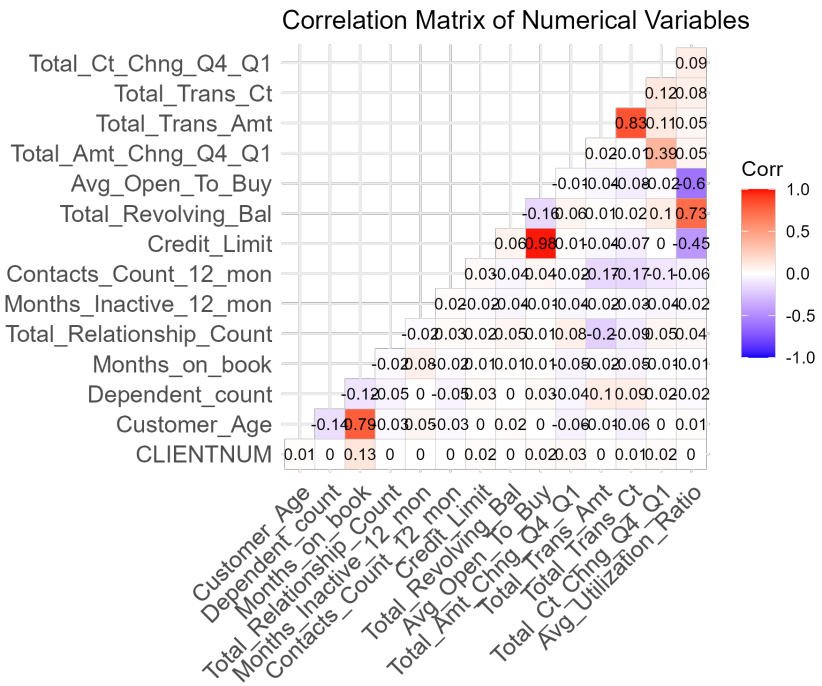

Figure 7: Correlation Matrix of all variables

B Appendix B

B.1 Clustering

B.1.1 Cluster Visualisation

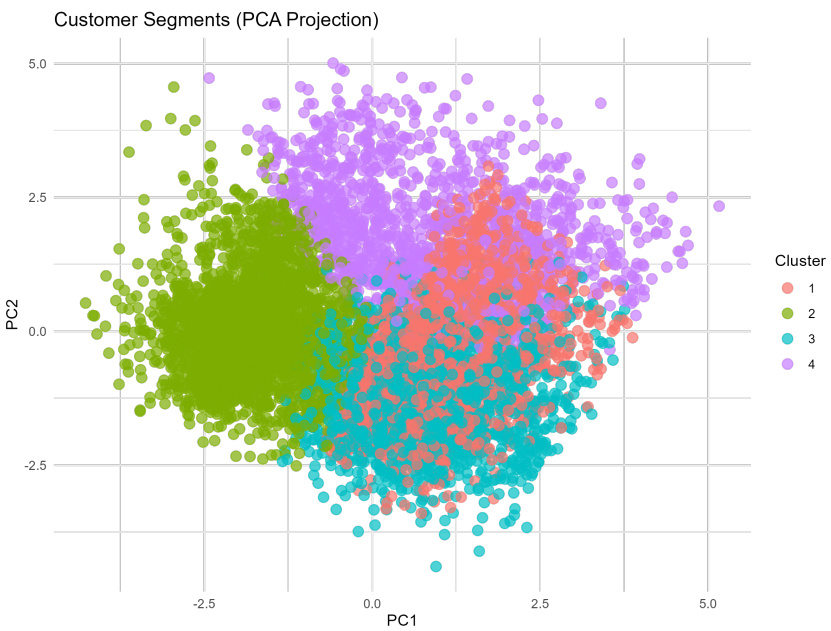

Figure 8: Cluster Scatter Plot with PCA

B.1.2 Cluster Profile Attributes

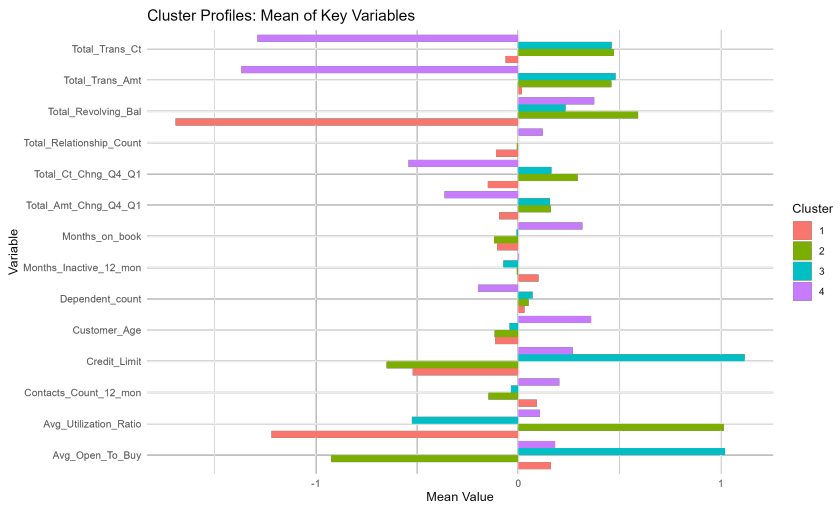

Figure 9: Mean values for each cluster

B.1.3 Hierarchical Clustering Dendogram

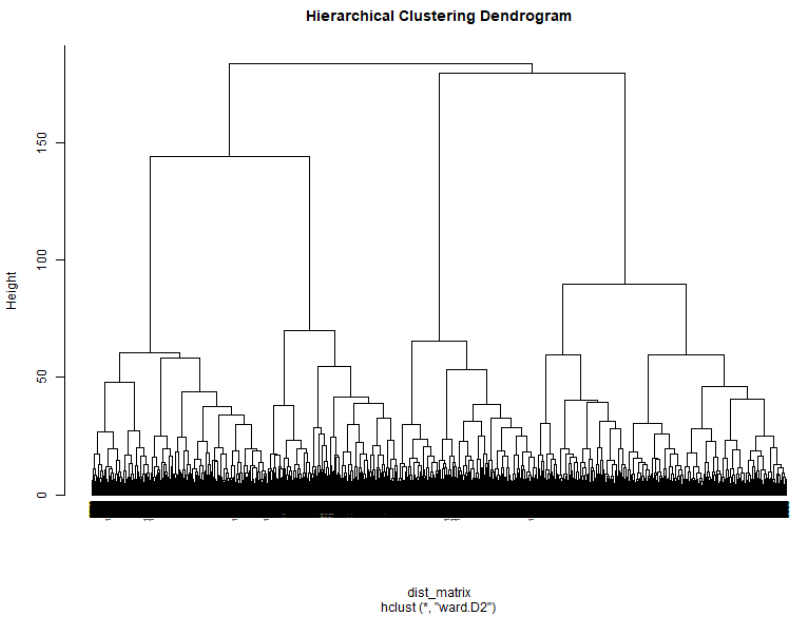

Figure 10: Hierarchical Clustering Dendogram

B.2 Regression

B.2.1 Partial Dependence Plots

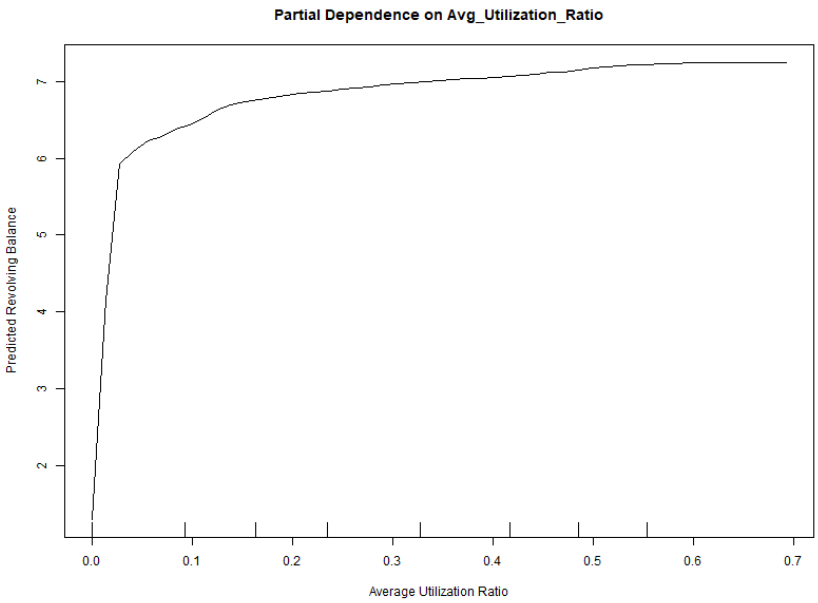

Figure 11: Partial Dependence Plot for Average Utilisation Ratio

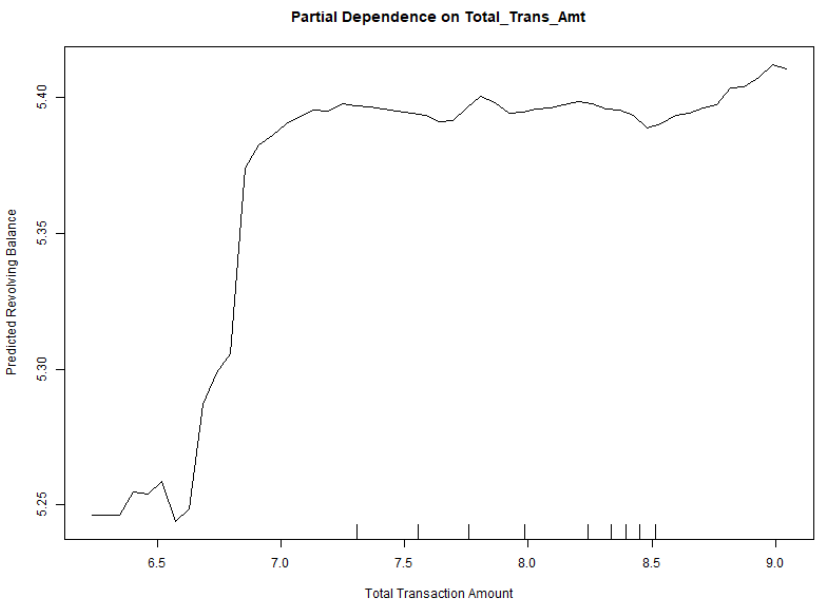

Figure 12: Partial Dependence Plot for Total Transaction Amount

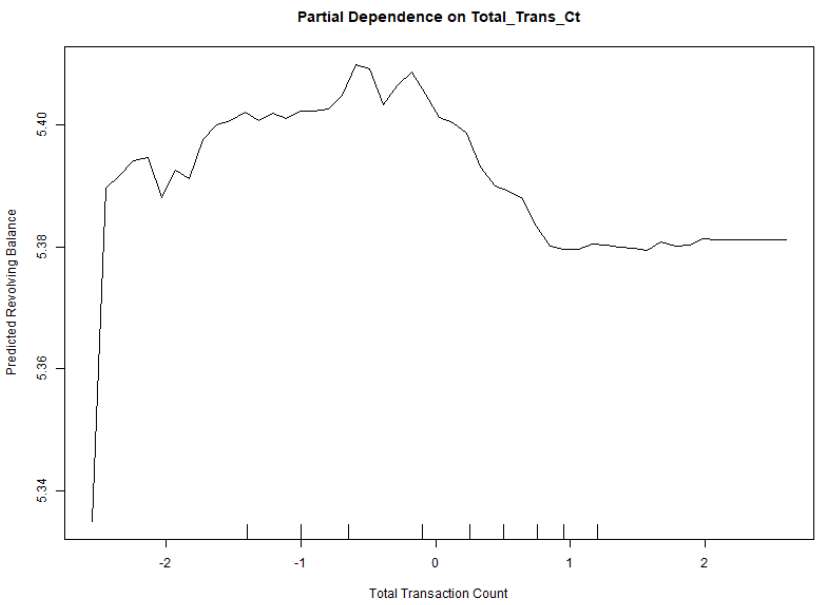

Figure 13: Partial Dependence Plot for Total Transaction Count

B.3 Classification

B.3.1 Partial Dependence Plots

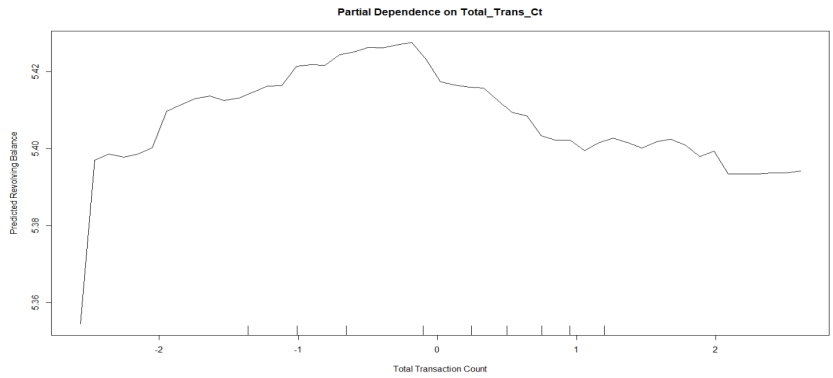

Figure 14: Partial Dependence Plot for Total Transaction Count

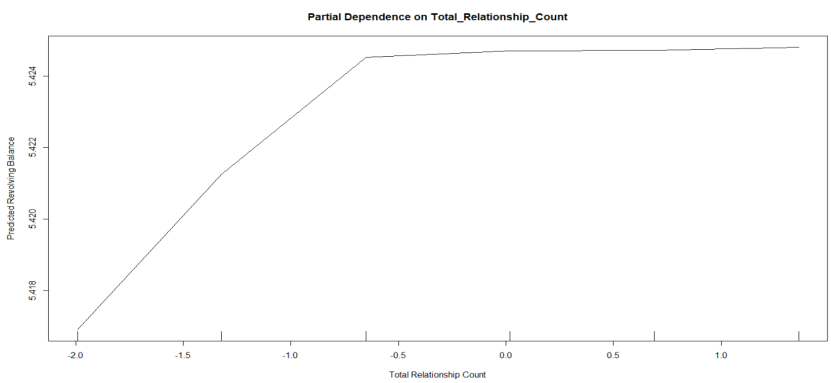

Figure 15: Partial Dependence Plot for Total Relationship Count